Multiple Self-Healing Squamous Epithelioma (MSSE / Ferguson-Smith Disease): A Comprehensive Disease Characteristics Report

Disease: Multiple Self-Healing Squamous Epithelioma (MSSE) **Causal gene: TGFBR1** (TGF-β receptor type 1) **MONDO:** MONDO:0007566 · **OMIM:** 132800 · **Orphanet:** 65748 · **NCIT:** C4461 · **DOID:** 5585 · **UMLS:** C0546476 · **MeSH:** C536150 · **SNOMED CT:** 254659009 · **MedGen:** 154270
Category: Mendelian (autosomal dominant)

Summary

Multiple Self-Healing Squamous Epithelioma (MSSE), also known as **Ferguson-Smith disease**, is a rare autosomal-dominant hereditary skin-cancer syndrome caused by germline **loss-of-function (LOF) variants in TGFBR1**, the gene encoding **transforming growth factor-β receptor type 1**. In this disease TGFBR1 behaves as a classic tumour-suppressor gene: affected individuals carry one defective germline allele, and tumours additionally show somatic loss of heterozygosity (LOH) with loss of the remaining wild-type allele. The hallmark of the disease is the recurrent eruption of multiple skin tumours that histologically resemble well-differentiated squamous cell carcinoma (SCC) / keratoacanthoma (KA): these grow rapidly over weeks, invade locally, and then **spontaneously regress** after several months, leaving deep, disfiguring facial scars and shallower limb scars. Lesions predominate on sun-exposed skin.

A central and scientifically interesting feature of TGFBR1 biology is that the same gene produces **two distinct allelic disorders**. MSSE results from **loss-of-function** variants (nonsense, frameshift, splice-site, and extracellular ligand-binding-domain missense changes), whereas **Loeys-Dietz syndrome (LDS)** — a connective-tissue/aortopathy disorder — results from **dominant-negative missense** variants in the serine/threonine kinase domain. The divergence is attributable to mutation nature and location, haploinsufficiency versus dominant-negative mechanisms, hypomorphicity, and modifier genes/environment. Haplotype

studies further indicate MSSE has a **digenic/multilocus aetiology**, requiring a permissive second locus within a chromosome 9q22.3 haplotype adjacent to TGFBR1, which likely explains why some TGFBR1-LOF carriers develop MSSE while others (or those with kinase-domain missense variants) manifest LDS or overlapping phenotypes.

Mechanistically, loss of TGF-β/SMAD2/3 growth-suppressive signalling releases keratinocytes from cell-cycle control, predisposing sun-exposed skin to invasive KA/SCC-like tumours; the characteristic **spontaneous regression** is immune/inflammation-mediated, associated with a cytotoxic CD8+ T-lymphocyte response and an infiltrate rich in **plasmacytoid dendritic cells**, with apoptosis executed via Fas/Fas-ligand and granzyme-B/perforin pathways. Prognosis for survival is good (no cardiovascular risk, unlike LDS); morbidity derives from recurrent lesions and disfiguring scarring. Management uses oral **acitretin**, topical **5-fluorouracil**, intralesional **triamcinolone**, surgical excision, and (in aggressive cases) EGFR-targeted therapy — while **radiotherapy is strictly avoided** because it severely exacerbates the disease.

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1. Disease Information

MSSE is a rare, inherited skin-cancer predisposition syndrome in which affected individuals develop multiple, recurrent, histologically malignant-appearing cutaneous tumours that undergo spontaneous regression (PMID: 22533930). It was first described in Scottish families by Ferguson-Smith and is therefore eponymously named.

Key identifiers (verified via Monarch/MONDO):

Resource	Identifier
MONDO	MONDO:0007566
OMIM (disease)	132800
Orphanet	65748
NCIT	C4461
DOID	5585
UMLS	C0546476
MeSH	C536150
SNOMED CT	254659009
MedGen	154270
Gene (TGFBR1)	HGNC:11772; Ensembl ENSG00000106799; OMIM 190181; chr9:99,104,038–99,154,192 (GRCh38); cytoband 9q22.33

Synonyms: Ferguson-Smith syndrome / disease / tumour; MSSE; familial primary self-healing squamous epithelioma of the skin, Ferguson-Smith type; multiple keratoacanthoma, Ferguson-Smith type; multiple self-healing epithelioma of Ferguson-Smith.

Information source: Information is derived almost entirely from **aggregated disease-level resources** — OMIM, Orphanet, and primary literature based on a small worldwide set of affected families (18 families / 67 individuals in the defining mutation study) — rather than from large EHR/population datasets, reflecting the condition's ultra-rarity.

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2. Etiology

Primary cause — genetic. MSSE is caused by germline **loss-of-function monoallelic mutations in TGFBR1** acting as a tumour-suppressor gene. High-throughput sequencing of the chromosome 9 disease locus followed by dideoxy sequencing identified **11 distinct monoallelic TGFBR1 mutations in 18 affected families**, firmly establishing TGFBR1 as

causative (PMID: 21358634). The variant spectrum — mutations in the extracellular ligand-binding domain plus a series of truncating mutations in the kinase domain — indicates a clear genotype–phenotype correlation with loss of function.

> *"Subsequent dideoxy sequencing of TGFBR1 identified 11 distinct monoallelic mutations in 18 affected families, firmly establishing TGFBR1 as the causative gene."* — PMID: 21358634

Two-hit tumour-suppressor mechanism. Molecular analysis of MSSE tumours demonstrated **loss of heterozygosity of the MSSE region with loss of the normal allele**, the first evidence that MSSE acts as a tumour suppressor (PMID: 16170343).

> *"We also report the first molecular analysis of MSSE tumours showing loss of heterozygosity of the MSSE region, with loss of the normal allele, providing the first evidence that MSSE is a tumour suppressor gene."* — PMID: 16170343

Genetic risk factors / digenic aetiology. TGFBR1 was initially *excluded* because it lies outside the extensive chromosome 9 haplotype shared by Scottish families; evidence now supports a **digenic/multilocus model** requiring a second linked locus within the chr9q22.3 Scottish haplotype adjacent to TGFBR1 (PMID: 24747516; PMID: 23358096; PMID: 33256177).

> *"MSSE can now be regarded as a digenic/multilocus disease in view of the evidence of a second linked locus necessary for pathogenesis located within the Scottish haplotype."* — PMID: 24747516

Environmental risk factors. Lesions predominate on sun-exposed skin (face, limbs), implicating **ultraviolet (UV) radiation**. Because MSSE lesions are keratoacanthomas, the broader KA risk factors apply: UV exposure, chemical carcinogens, chemical peels, trauma, thermal burns, scarring skin conditions, and immunosuppression (PMID: 16683389).

> *"In addition to ultraviolet exposure, KAs have also been associated with chemical carcinogens, chemical peels, genetic factors, chronic skin conditions that produce scarring, trauma and thermal burns."* — PMID: 16683389

Gene–environment interactions. Radiotherapy provokes severe exacerbation with eruptive new SCCs/KAs (PMID: 19874360; PMID: 28101021), and MSSE has arisen in a patient under systemic immunosuppression for rheumatoid arthritis (PMID: 37625788) — both demonstrating that environmental/iatrogenic factors interact with the germline TGFBR1 defect to modulate lesion burden.

Protective factors. No specific genetic or dietary protective factors are established for MSSE. By inference, **sun protection** and **avoidance of radiotherapy and skin trauma** reduce lesion induction. No protective modifier alleles have been reported.

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3. Phenotypes

The defining phenotype is **self-healing squamous epithelioma** (HPO canonical term **HP:0034720**). Lesions are **clinical signs / physical manifestations**: multiple tumours resembling crateriform squamous carcinomas that are locally invasive but regress spontaneously after several months, leaving deep disfiguring facial scars and shallower limb scars (PMID: 24747516).

> *"multiple tumours resembling crateriform squamous carcinomas are locally invasive but regress spontaneously after several months, leaving deep disfiguring facial scars and shallower scars on the limbs"* — PMID: 24747516

Phenotype	HPO term	Type	Onset	Severity / course	Frequency
Self-healing squamous epithelioma (KA/SCC-like tumours)	HP:0034720	Clinical sign / neoplasm	Variable: adolescence → late adulthood (7th decade cases reported)	Rapid growth (weeks) → local invasion → spontaneous regression (months); episodic/ recurrent	Defining feature (≈100%)
Facial scarring (disfiguring)	HP:0000987 (abnormal skin morphology)	Physical manifestation	Follows lesion regression	Deep, disfiguring; progressive scar accumulation	High
Limb/ extremity tumours & scars	—	Physical manifestation	Variable	Shallower scars	Common
Predilection for sun-exposed sites	—	Distribution feature	—	—	Characteristic

Age of onset is variable and age-dependent in expressivity; a confirmed late-onset case presented in a woman in her 60s (PMID: 28101021).

> *"We report the case of a woman in her 60s with confirmed multiple self-healing squamous epitheliomas (MSSE) Ferguson-Smith."* — PMID: 28101021

Quality-of-life impact: The principal burden is cosmetic and psychosocial — **recurrent disfiguring facial scarring** — rather than mortality. Lesions can occasionally ulcerate and behave aggressively, requiring surgery and reconstruction (PMID: 38541097).

Note on allelic overlap: Several HPO terms annotated to the OMIM/MONDO entry — Arterial tortuosity (HP:0005116), Ascending tubular aorta aneurysm (HP:0004970), Talipes equinovarus (HP:0001762), Bifid uvula (HP:0000193), Hypertelorism (HP:0000316) — reflect **allelic Loeys-Dietz overlap** rather than classic MSSE features and should not be attributed to typical MSSE.

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4. Genetic / Molecular Information

Causal gene: TGFBR1 (HGNC:11772; gene OMIM 190181; Ensembl ENSG00000106799; chr9q22.33).

Pathogenic variants. MSSE alleles are **germline, private (family-specific) mutations**: nonsense, frameshift, splice-site, and extracellular ligand-binding-domain missense changes, all producing **loss of function** (PMID: 21358634).

> *"The nature of the sequence variants, which include mutations in the extracellular ligand-binding domain and a series of truncating mutations in the kinase domain, indicates a clear genotype-phenotype correlation between loss-of-function TGFBR1 mutations and MSSE."* — PMID: 21358634

Allelic disorder contrast (MSSE vs LDS):

Feature	MSSE	Loeys-Dietz syndrome (LDS)
Variant class	Nonsense, frameshift, splice-site, extracellular-domain missense	Kinase (STK)-domain missense
Mechanism	Loss of function / haploinsufficiency	Dominant-negative
System	Skin (tumour suppressor)	Connective tissue / aorta
Cardiovascular risk	None	Aortic aneurysm/ dissection

> *"Conversely, nonsense, frameshift, or specific missense changes in the ligand-binding extracellular domain cause multiple self-healing squamous epithelioma (MSSE) lacking the cardiovascular phenotype."* — PMID: 36599937

Functional splicing assays confirm the mechanism: an MSSE intron-4 splice variant (c.806-2A>C) produces an out-of-frame/truncating transcript (p.Asn270Thrfs*8 → LOF), whereas a comparable LDS variant yields in-frame dominant-negative products (PMID: 29706644).

> *"Our results support the previously proposed but not yet approved mechanism that dominant-negative and truncating variants in STK domain induce LDS and MSSE, respectively."* — PMID: 29706644

Rare patients carrying **kinase-domain missense** variants display **overlapping MSSE + LDS** features (e.g., p.Gly222Arg; PMID: 30513140, PMID: 38112607, PMID: 39933850). A broad review attributes phenotypic divergence to mutation nature/location, hypomorphicity, haploinsufficiency vs dominant-negative effects, and modifier genes/environment (PMID: 40612107).

> *"Multiple Self-healing Squamous Epithelioma (MSSE), a rare familial skin cancer, is linked to mutations in these genes."* — PMID: 40612107

Constraint metrics (gnomAD v4, GRCh38). TGFBR1 is **intolerant to loss-of-function and missense variation**, concordant with the MSSE LOF mechanism: pLI = 0.91; LOEUF = 0.53 (observed/expected LoF = 20/55.1; oe_lof = 0.36); LoF Z = 4.01; missense Z = 3.99.

Somatic vs germline. The causal variant is **germline**; tumours acquire a **somatic second hit (LOH)** removing the wild-type allele (PMID: 16170343).

Allele frequency. MSSE alleles are absent from population databases (private family mutations), supporting ACMG criterion PM2 (absence from gnomAD) alongside PVS1 for null variants.

ClinVar. NCBI ClinVar (queried Nov 2026) holds ~1,306 TGFBR1 variant records (~594 pathogenic), of which ~50–81 are explicitly annotated to "multiple self-healing squamous epithelioma," supporting single-gene diagnostic testing (finding F015). Variant classification follows ACMG via LOF criteria (PVS1 for null variants + PM2 for absence from gnomAD).

Modifier genes / chromosomal abnormalities. A permissive **second locus within chr9q22.3** acts as a modifier/necessary co-factor (PMID: 24747516; PMID: 23358096). No recurrent large-scale aneuploidies or translocations are reported. No MSSE-specific epigenetic signature has been characterized.

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5. Environmental Information

- **Environmental factors:** UV radiation (sun-exposed predilection), chemical carcinogens, chemical peels, trauma, thermal burns, and scarring skin conditions — all recognized KA inducers (PMID: 16683389).
- **Iatrogenic: Radiotherapy** strongly exacerbates MSSE, causing extensive eruptive mid-facial SCCs/KAs (PMID: 19874360).

> *"extensive, eruptive mid-facial squamous cell carcinomas (SCCs) and keratoacanthomas (KAs) consequent to radiotherapy"* — PMID: 19874360

- **Immune status:** Immunosuppression (post-transplant, and a reported RA immunosuppression case) promotes KA/MSSE lesions (PMID: 16683389; PMID: 37625788).

> *"Immunosuppressed patients, especially after transplantation, also develop KAs."* — PMID: 16683389

- **Infectious agents:** Not an established cause of MSSE. (HPV is linked to some sporadic KAs but is not implicated in germline MSSE.)

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6. Mechanism / Pathophysiology

Ordered causal chain

1. A **germline loss-of-function TGFBR1 variant** (nonsense/frameshift/splice-site/extracellular-domain missense) **reduces functional TGF-β type I receptor** → TGFBR1 haploinsufficiency (demonstrated: LOF spectrum, gnomAD constraint).

2. In a skin keratinocyte, a **somatic second hit (LOH)** removes the wild-type TGFBR1 allele → **complete loss of TGF-β receptor signalling** in that clone (demonstrated: tumour LOH, PMID: 16170343).
3. Loss of receptor signalling **abolishes TGF-β/SMAD2/3 activation** → loss of SMAD-driven induction of cell-cycle inhibitors (p21^WAF1^, p12^CDK2-AP1^) (demonstrated in keratinocytes, PMID: 17217620).
4. Loss of growth suppression **releases keratinocytes from cell-cycle control** → clonal proliferation, invasion (inferred for MSSE; supported by cSCC data, PMID: 29581863).
5. On **sun-exposed skin (UV co-factor)** this yields a rapidly growing, locally invasive **keratoacanthoma/SCC-like tumour** (demonstrated clinically, PMID: 24747516).
6. **Branch — spontaneous regression:** the tumour triggers an **immune/inflammatory response** → infiltration by cytotoxic CD8+ T lymphocytes (with CD4+ helpers) and **plasmacytoid dendritic cells** → tumour-cell **apoptosis via Fas/FasL and granzyme-B/perforin** → spontaneous regression leaving a scar (demonstrated: pDC infiltrate in regressed MSSE KA, PMID: 28101021; CTL/granzyme-B mechanism in KA, PMID: 16497444).

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Germline TGFBR1 LOF (one allele)
        | haploinsufficiency
        ▼
Somatic LOH (loss of wild-type allele in keratinocyte)
        | complete loss of TGF-β receptor
        ▼
No SMAD2/3 activation ──► ↓ p21^WAF1^, ↓ p12^CDK2-AP1^
        | loss of growth suppression       (+ UV co-factor)
        ▼
Keratoacanthoma / SCC-like tumour (rapid growth, local invasion)
        |
        ├──► Immune recognition: CD8+ CTL + pDC infiltrate
        |        | Fas/FasL + granzyme-B/perforin apoptosis
        |        ▼
        |   SPONTANEOUS REGRESSION → disfiguring scar
        |
        └──► (iatrogenic branch) Radiotherapy → severe exacerbation, eruptive new lesions
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Molecular pathways and supporting detail

- **TGF-β/SMAD signalling (upstream driver).** TGF-β1 treatment of normal keratinocytes suppresses growth via upregulation of p-SMAD2, p12^CDK2-AP1^ and p21^WAF1^; these effects are lost in SCC lines with disrupted TGF-β–SMAD signalling (PMID: 17217620). Inactivating mutations in TGF-β receptors are genetic drivers of sporadic and murine cSCC,

and reduced SMAD2/3 activation independently predicts increased cSCC depth (PMID: 29581863).

> *"Inactivating mutations in transforming growth factor beta (TGF-β) receptors have been identified amongst genetic drivers of sporadic tumours and murine models of cSCC, suggesting a tumour suppressor function for TGF-β in normal skin"* — PMID: 29581863
>
> *"TGF-beta1 treatment of normal keratinocytes suppressed cell growth with upregulation of p-Smad2, p12(CDK2-AP1), and p21(WAF1) expression"* — PMID: 17217620

- **Cellular processes:** cell-cycle dysregulation (loss of p21/p12 checkpoint induction), epithelial proliferation/invasion, and immune-mediated apoptosis during regression.
- **Immune involvement (downstream, regression):** cytotoxic CD8+ T-lymphocyte response with CD4+ helpers; plasmacytoid dendritic cell infiltrate; Fas/FasL and granzyme-B/perforin effector pathways (PMID: 16497444; PMID: 28101021).

> *"the tumor regression is dependent mainly on the immune response mediated by cytotoxic T lymphocytes (CD8+), together with helper T cells (CD4+)"* — PMID: 16497444
>
> *"revealed an increased inflammatory infiltrate with numerous plasmacytoid dendritic cells in the regressed KA in comparison to the mature one, speaking in favor of an inflammation-mediated regression process"* — PMID: 28101021

Suggested ontology terms: GO:0007179 (transforming growth factor beta receptor signaling pathway); GO:0008285 (negative regulation of cell population proliferation); GO:0007049 (cell cycle); GO:0006915 (apoptotic process); GO:0006954 (inflammatory response). Cell types: CL:0000312 (keratinocyte); CL:0000784 (plasmacytoid dendritic cell); CL:0000625 (CD8-positive, alpha-beta T cell). Chemicals (CHEBI): CHEBI:63631 (acitretin), CHEBI:46345 (5-fluorouracil).

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7. Anatomical Structures Affected

- **Organ level: Skin** is the sole primary organ affected (integumentary system). No secondary organ involvement in classic MSSE (cardiovascular involvement belongs to the allelic LDS phenotype, not MSSE).

- **Tissue/cell level: Epithelial tissue** — specifically the stratified squamous epidermis; the targeted cell population is the **keratinocyte** (CL:0000312). Desmoglein (Dsg1/Dsg3) staining is uniformly preserved throughout the non-keratinized layers of MSSE tumours — similar to actinic keratoacanthoma and distinct from invasive SCC (PMID: 9705163).

> *"All eight tumors showed uniform pericellular Dsg staining throughout the nonkeratinized layers of the neoplastic epithelium. This pattern is entirely similar to that observed in actinic KA"* — PMID: 9705163

- **Subcellular level:** TGFBR1 is a transmembrane receptor (plasma membrane, GO:0005886); SMAD effectors shuttle to the nucleus (GO:0005634).
- **Localization (UBERON):** skin of face (UBERON:0008200 head skin), skin of limb (UBERON:0002097 skin of body; limb skin), with **predilection for sun-exposed sites**. Distribution is typically **bilateral/multifocal** (multiple recurrent lesions), not strictly lateralized.

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8. Temporal Development

- **Onset:** Variable and **age-dependent in expressivity** — reported from adolescence through late adulthood, including a confirmed presentation in the 7th decade (PMID: 28101021). Onset of individual lesions is **acute/subacute** (rapid growth over weeks).
- **Progression:** Individual lesions follow a stereotyped **episodic course** — rapid growth → local invasion → spontaneous regression over several months — leaving a scar (PMID: 24747516). The disease as a whole is **chronic and recurrent/lifelong**, with new crops of lesions and accumulating scars.
- **Remission patterns:** Spontaneous regression of each lesion is the defining pattern; treatment (acitretin, intralesional steroid) can accelerate/limit lesions.
- **Critical periods / caution: Radiotherapy** at any time can trigger severe eruptive exacerbation and should be avoided (PMID: 19874360; PMID: 28101021).

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9. Inheritance and Population

- **Inheritance: Autosomal dominant** (PMID: 22533930; PMID: 21358634).

> *"Ferguson-Smith disease (MSSE) is an autosomal dominant inherited disease with multiple, recurrent, histologically malignant tumours that undergo spontaneous regression."* — PMID: 22533930

- **Epidemiology: Ultra-rare.** The defining mutation study identified TGFBR1 mutations in only **18 families (67 individuals)** worldwide (PMID: 21358634; PMID: 22758773). Precise prevalence/incidence are not established (Orphanet:65748).

> *"mutations of the TGFBR1 gene were demonstrated in 18 families (67 individuals) with Ferguson-Smith disease"* — PMID: 22758773

- **Penetrance / expressivity:** High penetrance with **variable, age-dependent expressivity** (onset and lesion burden vary). A permissive **9q22.3 modifier locus** contributes to whether a TGFBR1-LOF carrier manifests MSSE (digenic model) (PMID: 24747516).
- **Founder effects / population:** Originally described in **Scottish families** with a founder-like haplotype, but subsequently identified across **multiple ethnic groups with distinct at-risk haplotypes**, arguing against a single founder mutation and suggesting MSSE may be more common than originally thought (PMID: 17554363).

> *"The haplotypes for polymorphic markers segregating with MSSE in non-Scottish and Scottish families differ, suggesting that MSSE is not caused by a founder mutation and might be more common than originally thought."* — PMID: 17554363

- **Sex ratio / age distribution:** No strong sex predilection established; onset spans adolescence to late adulthood. Consanguinity is not a factor (dominant). Germline mosaicism and carrier frequency are not specifically characterized.

10. Diagnostics

- **Clinical diagnosis:** Recognition of multiple rapidly growing, crateriform, self-regressing squamous tumours on sun-exposed skin in a patient with a positive family history.
- **Histopathology / IHC:** Lesions resemble well-differentiated SCC/keratoacanthoma. **Desmoglein (Dsg) immunostaining is uniformly preserved**, matching actinic KA and distinguishing MSSE lesions from invasive SCC (PMID: 9705163) — a useful adjunct, since routine histology cannot reliably separate KA from SCC (PMID: 22758773).
- **Genetic testing: Single-gene testing of TGFBR1** is the confirmatory approach; WES/WGS or panels including TGFBR1 are alternatives. ClinVar holds curated MSSE-associated TGFBR1 records (germline, private, pathogenic LOF alleles) (finding F015). CMA/karyotyping/FISH/mtDNA/repeat-expansion testing are **not applicable** (single-gene LOF point mutations).
- **Differential diagnosis:** sporadic keratoacanthoma, invasive cutaneous SCC, multiple KA syndromes (e.g., Grzybowski, Witten-Zak), Muir-Torre syndrome, and the allelic **Loeys-Dietz syndrome** (distinguished by connective-tissue/aortic features and kinase-domain missense variants).
- **Omics / imaging / electrophysiology:** No disease-specific biomarker, imaging, or electrophysiology test; diagnosis is clinical + histological + molecular.
- **Screening:** Cascade genetic testing of at-risk relatives in known families; no population-level newborn/carrier screening (ultra-rare, dominant).

11. Outcome / Prognosis

- **Survival / mortality: Excellent survival.** Unlike the allelic LDS, MSSE carries **no cardiovascular risk**; lesions regress spontaneously and the disease is not life-threatening.
- **Morbidity / function:** Principal burden is **recurrent disfiguring scarring** (deep facial scars, shallower limb scars) and psychosocial impact (PMID: 24747516). Occasional lesions ulcerate or behave aggressively, requiring excision and reconstruction (PMID: 38541097).
- **Disease course:** Chronic, recurrent, lifelong with spontaneous per-lesion remission.
- **Prognostic factors:** Lesion frequency/burden and exposure to exacerbating factors (radiotherapy, immunosuppression, UV). No validated molecular prognostic biomarker beyond the causal genotype.

12. Treatment

No curative therapy exists; management controls lesion burden and treats aggressive lesions.

Modality	Agent / approach	Evidence	NCIT suggestion
Systemic retinoid	**Oral acitretin** (treatment of choice; e.g., 0.5 mg/kg/day)	PMID: 19874360; PMID: 38541097	NCIT:C1878 (Acitretin)
Topical chemotherapy	**5-Fluorouracil 5%** (often with acitretin)	PMID: 38541097	NCIT:C505 (Fluorouracil)
Intralesional corticosteroid	**Triamcinolone acetonide** (acts on proliferation phase; good for multiple lesions)	PMID: 28101021	NCIT:C780 (Triamcinolone)
Surgery	Excision + plastic reconstruction for ulcerated/ invasive lesions	PMID: 38541097	NCIT:C15329 (Surgical Procedure)
EGFR-targeted + chemo	**Cetuximab** + **cisplatin** for multiple aggressive KAs	PMID: 33871373	NCIT:C1723 (Cetuximab); NCIT:C376 (Cisplatin)
Avoid	**Radiotherapy** — severe exacerbation	PMID: 19874360; PMID: 28101021	—

> *"We found oral acitretin to be the treatment of choice."* — PMID: 19874360
>
> *"Multiple keratoacanthomas on her extremities were successfully treated with oral acitretin (0.5 mg/kg/day) in combination with topical Fluorouracil (5-FU) 5%, while excision and plastic surgery restoration were performed"* — PMID: 38541097
>
> *"Intralesional triamcinolone acetate injection is a safe and easy to use method also effective for multiple lesions."* — PMID: 28101021

Pharmacogenomics / advanced therapeutics: No gene/cell/RNA therapy exists for MSSE. Targeted anti-EGFR therapy is used empirically for aggressive lesions. No MSSE-specific registered clinical-trial programs (NCT) are established.

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13. Prevention

- **Primary prevention:** No way to prevent the inherited mutation; **sun protection** and **avoidance of skin trauma, chemical peels, and radiotherapy** reduce lesion induction (inferred from KA risk factors, PMID: 16683389; radiotherapy caution, PMID: 19874360).
- **Secondary prevention:** Regular dermatological surveillance for early identification/treatment of lesions; early retinoid therapy to limit crops.
- **Tertiary prevention:** Timely excision of ulcerated/aggressive lesions; careful management to minimize scarring.
- **Genetic counseling:** Autosomal-dominant counseling (50% offspring risk); cascade genetic testing of at-risk relatives; prenatal/preimplantation testing is theoretically possible given known family variants.
- **Immunization / public health / prophylaxis:** Not applicable (non-infectious Mendelian disease).

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14. Other Species / Natural Disease

- **Taxonomy:** MSSE is described in **humans** (NCBI Taxon 9606). No naturally occurring MSSE is documented in companion animals or wildlife (not in OMIA as a recognized natural homolog for this specific syndrome).
- **Orthologous gene:** *Tgfbr1* is highly conserved — mouse *Tgfbr1* (NCBI Gene 21812), rat *Tgfbr1* — enabling model-organism study.
- **Comparative biology / evolutionary conservation:** The TGF-β/SMAD tumour-suppressor axis is deeply conserved; its loss produces cutaneous SCC in mice (see Section 15), supporting conservation of the disease mechanism even where the exact human syndrome is not naturally reproduced.
- **Zoonotic potential:** None (non-infectious).

15. Model Organisms

No single model fully recapitulates MSSE (germline TGFBR1 LOF + 9q modifier + spontaneous regression). However, **mouse models of downstream TGF-β/SMAD effector loss** strongly support the tumour-suppressor mechanism:

Model	Genotype	Phenotype	Evidence
Epidermis-specific Smad4 KO	*Smad4*^-/-^ keratinocytes	Spontaneous malignant cutaneous **SCC** (3–13 months); blocked TGF-β/BMP signalling, loss of p21, Pten inactivation, AKT/cyclin D1 activation	PMID: 16170355; PMID: 18618014
Keratinocyte-specific Smad2 KO	*Smad2*^-/-^ keratinocytes	Accelerated chemically-induced skin tumours; malignant progression with EMT via Snail	PMID: 18618014

> *"all MT mice developed spontaneous malignant skin tumors from 3 months to 13 months of age. The majority of tumors are malignant squamous cell carcinomas"* — PMID: 16170355
>
> *"Smad4-/- mouse epidermis develops spontaneous skin squamous cell carcinomas (SCCs)"* — PMID: 18618014

Recapitulation & limitations: These models confirm that disrupting the TGF-β/SMAD axis in keratinocytes is sufficient to cause cutaneous SCC — validating the upstream mechanism — but they **do not reproduce the self-healing/spontaneous-regression phenotype** or the digenic 9q modifier requirement central to human MSSE. **Resources:** MGI (mouse), RGD (rat). In-vitro keratinocyte and SCC cell-line models demonstrate the TGF-β1 → p-SMAD2 → p21/p12 growth-suppression axis (PMID: 17217620).

Mechanistic Model / Interpretation

MSSE is best understood as a **two-hit tumour-suppressor disease with an immune "brake" that forces regression**. The germline LOF TGFBR1 allele primes keratinocytes; a somatic second hit (LOH) in sun-exposed skin eliminates residual TGF-β signalling, releasing SMAD2/3-dependent cell-cycle control (↓p21, ↓p12) and allowing a rapidly growing, locally invasive KA/SCC-like tumour. Crucially, these tumours retain features of keratoacanthoma (preserved desmoglein, well-differentiated histology) and are **highly immunogenic**, so a cytotoxic CD8+ T-cell / plasmacytoid-dendritic-cell response drives Fas/FasL- and granzyme-B-mediated apoptosis and spontaneous regression — leaving a scar. The genotype–phenotype logic is the report's most elegant insight: **the same gene causes two opposite-system diseases** depending on variant mechanism (LOF/haploinsufficiency → skin tumours in MSSE; dominant-negative kinase-domain missense → connective-tissue/aortopathy in LDS), modulated by a permissive 9q22.3 modifier locus and environment (UV, radiotherapy, immunosuppression).

Evidence Base (key literature)

PMID	Role
21358634	**Landmark:** establishes TGFBR1 LOF mutation spectrum as causal in 18 families
16170343	LOH with loss of wild-type allele → tumour-suppressor proof
24747516	Digenic/multilocus model; clinical course; scarring
17554363	Multiple ethnic groups; not a single founder mutation
36599937 / 29706644	MSSE (LOF) vs LDS (dominant-negative) genotype–phenotype
40612107	Review: hypomorphicity, haploinsufficiency, modifiers
29581863 / 17217620	TGF-β/SMAD2/3 keratinocyte tumour suppression
16170355 / 18618014	Mouse Smad4/Smad2 KO → spontaneous cutaneous SCC
28101021 / 16497444	Immune-mediated regression (pDC; CTL/granzyme-B/Fas)
9705163	Desmoglein IHC distinguishes MSSE/KA from SCC
19874360 / 38541097 / 33871373	Treatment: acitretin, 5-FU, surgery, cetuximab; radiotherapy caution
22533930 / 22758773	AD inheritance; rarity (18 families / 67 individuals)

Limitations and Knowledge Gaps

- **Ultra-rarity:** Evidence derives from ~18 families; no formal prevalence/incidence, penetrance quantification, or sex-ratio data exist.
- **Unidentified second locus:** The digenic 9q22.3 modifier gene is inferred from haplotype data but **not molecularly identified**.
- **Regression mechanism:** Immune-mediated regression in MSSE is supported by a single detailed case (pDC infiltrate) plus KA analogy; the precise trigger and the role of restored vs residual TGF-β signalling remain uncharacterized.

- **No faithful animal model:** Existing mouse models capture SCC formation from TGF-β/SMAD loss but not the self-healing phenotype or 9q modifier.
- **No omics datasets:** No MSSE-specific transcriptomic/proteomic/metabolomic/epigenomic signatures are published.
- **Genotype–phenotype overlap:** Kinase-domain missense cases blur the MSSE/LDS boundary, complicating counseling.

Proposed Follow-up Experiments / Actions

1. **Identify the chr9q22.3 modifier locus** via deep sequencing and functional dissection of the at-risk haplotype shared by MSSE families, to explain MSSE-vs-LDS divergence.
2. **Single-cell / spatial transcriptomics of matched growing vs regressing MSSE lesions** to define the immune regression program (pDC, CD8+ CTL states, interferon signatures) and identify immunomodulatory therapeutic targets.
3. **Build a faithful mouse model** combining epidermal *Tgfbr1* haploinsufficiency with the candidate 9q modifier and UV exposure to recapitulate self-healing tumours.
4. **Systematic ClinVar/HGMD variant curation** of MSSE vs LDS alleles with functional (splicing/receptor-signalling) assays to refine ACMG classification and genotype–phenotype rules.
5. **Prospective registry** across the known families to quantify prevalence, penetrance, onset distribution, lesion burden, and treatment response (acitretin, 5-FU, intralesional steroid, cetuximab).
6. **Formalize management guidance** emphasizing sun protection and an explicit contraindication to radiotherapy.

Evidence source types: predominantly human clinical/genetic (case series, family linkage, ClinVar) and in-vitro keratinocyte/SCC studies; mechanistic support from mouse model organism studies; computational constraint metrics from gnomAD.
